Supplementary material

Closing the Loop in Retrieval-Augmented Generation:
A Systematic Survey of Adaptive Retrieval, Reasoning, and Resource-Aware Intelligence

Zafar Ali, Khan Muhammad, Adam A. Q. Mohammed, Guilin Qi,
Heiko Paulheim, and Pavlos Kefalas

S1 Search and screening record

All routes use the same publication-eligibility cutoff, 5 November 2026. The manuscript gives the search concept blocks and eligibility criteria. Table S1 reconciles the available screening counts; the selection audit documents their provenance. The retained records do not contain complete executed queries or reasons for every report not obtained.

Table S1: Selection counts by discovery route.

Stage	Database and citation	Targeted	Total
Identified records	5,130	89	5,219
Duplicates before screening	1,642	5	1,647
Records screened	3,488	84	3,572
Title/abstract/scope exclusions	3,215	4	3,219
Reports sought	273	80	353
Reports not retrieved	18	3	21
Full texts assessed	255	77	332
Full-text exclusions	184	17	201
Included analytical studies	71	60	131

The 21 reports not retrieved are distinct from the three reports assessed using supplied copies and excluded because public full-text availability could not be verified. The eligibility ledger records the candidate decisions it covers; it is not a log of all records screened.

S2 Included studies and reporting appraisal

Table S2 identifies each of the 131 included studies. The corpus and appraisal CSVs contain the corresponding titles, publication records, and codes. **Emph.**: T, triggering; R, routing; B, budget/evidence; S, stopping/sufficiency. Each study receives one primary emphasis even if it operates in several phases.

Appraisal: C, clearly reported or addressed; P, partially reported; NR, not clearly established. AD, adaptive decision; CO, controller observability; BL, baseline adequacy; AB, ablation quality; RR, resource reporting; EB, evaluation breadth; SR, statistical robustness; RP, reproducibility; FA, failure/robustness analysis; CE, claim–evidence alignment. The codes are not combined into a quality score.

Table S2: Included studies and their ten reporting-appraisal codes.

Study	Year	Emph.	AD	CO	BL	AB	RR	EB	SR	RP	FA	CE
FLARE [1]	2023	T	C	C	C	C	C	C	NR	C	P	C
IRCoT [2]	2023	R	C	C	C	P	C	C	NR	C	P	C
Iter-RetGen [3]	2023	R	C	C	C	P	P	C	NR	P	NR	C
Rewrite-Retrieve-Read [4]	2023	R	C	C	C	P	P	C	NR	P	NR	C
SKR [5]	2023	T	C	C	C	P	P	C	NR	P	P	C
Adaptive-RAG [6]	2024	R	C	C	C	C	C	C	P	C	P	C
Conversational Adaptive RAG [7]	2024	T	C	C	C	P	P	C	NR	P	P	C
DRAGIN [8]	2024	T	C	C	C	C	C	C	NR	C	C	C
EfficientRAG [9]	2024	B	C	C	C	C	C	C	NR	C	P	C
Query-Knowledge Relevance [10]	2024	S	C	C	P	NR	NR	P	NR	P	P	C
RE-RAG [11]	2024	B	C	C	C	C	P	C	NR	P	C	C
REAR [12]	2024	B	C	P	C	C	NR	C	NR	P	C	C
RetrievalQA [13]	2024	T	C	C	C	P	P	C	NR	P	NR	C
Self-RAG [14]	2024	T	C	C	C	C	C	C	P	C	C	C

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Table S2 continued

Study	Year	Emph.	AD	CO	BL	AB	RR	EB	SR	RP	FA	CE
Self-Route [15]	2024	R	C	C	C	P	P	C	NR	P	NR	C
UAR [16]	2024	T	C	C	C	P	P	C	NR	P	P	C
Adaptive Retrieval Comparison [17]	2025	T	C	C	P	NR	NR	C	NR	P	C	C
Adaptive-k [18]	2025	B	C	C	C	C	C	C	NR	P	P	C
AMBER [19]	2025	B	C	P	C	P	P	C	NR	P	P	P
AMRAF [20]	2025	T	C	C	C	C	C	C	NR	P	C	C
CtrlA [21]	2025	T	C	C	C	P	P	C	NR	P	P	C
DioR [22]	2025	T	C	C	C	C	P	C	NR	P	C	C
DynamicRAG [23]	2025	B	C	C	C	C	C	C	NR	C	P	C
EI-ARAG [24]	2025	T	C	C	C	C	C	C	NR	C	P	C
FiSKE [25]	2025	S	C	C	C	C	C	C	NR	C	C	C
IRAGKR [26]	2025	T	C	C	C	C	C	C	NR	P	C	C
LLM-Independent Adaptive RAG [27]	2025	T	C	C	C	P	P	C	NR	P	P	C
MBA-RAG [28]	2025	R	C	C	C	P	C	C	NR	P	P	C
Overlap Reduction [29]	2025	B	C	P	C	P	C	P	NR	P	NR	P
Q-DREAM [30]	2025	R	C	P	C	C	P	C	P	C	C	P
SeaKR [31]	2025	T	C	C	C	C	C	C	P	C	C	C
Search-o1 [32]	2025	R	C	C	C	C	C	C	NR	C	P	C
Search-R1 [33]	2025	R	C	C	C	C	C	C	P	C	P	C
SGIC [34]	2025	B	C	P	C	P	P	P	NR	P	P	C
Smart-Searcher [35]	2025	R	C	C	C	P	C	C	NR	P	NR	C
Sufficient Context [36]	2025	S	C	C	P	NR	NR	C	NR	P	C	C
UniRAG [37]	2025	R	C	C	C	P	P	C	NR	P	P	C
ACRA [38]	2026	R	C	C	C	C	P	C	NR	P	C	C
AdaCache [39]	2026	B	C	C	C	C	C	C	P	P	C	C
AdaKG-RAG [40]	2026	S	C	C	C	C	P	P	P	P	C	C
AED-RAG [41]	2026	B	C	C	C	C	P	C	NR	P	C	C
Agentic-R [42]	2026	R	C	C	C	C	C	C	P	C	C	C
AgenticTableRAG [43]	2026	R	C	C	C	C	C	C	C	C	C	C
AIR-RAG [44]	2026	R	C	C	C	C	P	C	P	P	C	C
ATEA [45]	2026	T	C	C	C	C	C	C	C	P	C	C
AutoSearch [46]	2026	S	C	C	C	C	C	C	P	C	C	C
BAR-RAG [47]	2026	B	C	C	C	C	C	C	NR	C	C	C
BOLT [48]	2026	R	C	C	C	C	C	C	NR	C	C	C
BRIEF-Pro [49]	2026	B	C	P	C	P	C	P	NR	P	NR	P
CIRAG [50]	2026	R	C	C	C	C	C	C	NR	P	C	C
ClauseRoute [51]	2026	B	C	C	C	C	C	C	P	C	C	C
CompSelect [52]	2026	B	C	C	C	C	C	C	P	C	C	C
CONFRAG [53]	2026	T	C	C	C	C	C	C	NR	P	C	C
CORAL [54]	2026	R	C	C	C	C	C	C	NR	P	C	C
Cost-aware depth [55]	2026	B	C	C	C	C	C	C	C	P	C	C
Cost-Aware Evidence Selection [56]	2026	B	C	C	C	C	C	C	C	C	C	C
CRAG [57]	2026	B	C	C	C	C	C	C	NR	P	P	P
D2-RAG [58]	2026	T	C	C	C	C	P	C	P	P	C	C
DA-RAG [59]	2026	B	C	C	C	C	C	P	P	C	C	C
DeepRAG [60]	2026	R	C	C	P	P	C	C	NR	NR	P	C
Deliberative Searcher [61]	2026	T	C	C	C	C	C	C	NR	P	C	C
Doc-REFRAG [62]	2026	B	C	C	C	C	C	C	NR	C	C	C
DORA [63]	2026	R	C	C	C	C	C	C	P	C	C	C
ETC [64]	2026	T	C	C	C	C	C	C	C	C	C	C
Failure is Feedback [65]	2026	R	C	C	C	C	C	C	P	C	C	C
Flow-RAG [66]	2026	B	C	C	C	C	C	P	P	C	C	C
FrugalRAG [67]	2026	S	C	C	C	C	C	C	C	C	C	C
G2-Reader [68]	2026	S	C	C	C	C	C	C	C	C	C	C
GAM-RAG [69]	2026	B	C	C	C	C	C	C	C	P	C	C
GRAD [70]	2026	B	C	P	C	C	NR	C	NR	C	NR	P
Graph-R1 [71]	2026	R	C	C	C	C	C	C	NR	C	C	C
GraphRAG-R1 [72]	2026	R	C	C	C	C	C	C	NR	C	P	C
GraphRAG-Router [73]	2026	R	C	C	C	C	C	C	NR	C	C	C
GRIP [74]	2026	R	C	C	C	C	C	C	P	C	P	C
HALT [75]	2026	S	C	C	C	C	C	C	C	C	C	C
HiKEY [76]	2026	R	C	C	C	C	C	P	NR	P	P	C
INKER [77]	2026	T	C	C	C	C	C	C	P	P	C	C

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Table S2 continued

Study	Year	Emph.	AD	CO	BL	AB	RR	EB	SR	RP	FA	CE
InnerRAG [78]	2026	B	C	C	C	C	C	C	C	P	C	C
Internal Confidence [79]	2026	T	C	C	C	C	C	C	NR	C	C	C
IterCOMP [80]	2026	B	C	C	C	P	C	P	NR	P	P	C
LAD-RAG [81]	2026	R	C	C	C	C	C	C	NR	P	P	C
LCR [82]	2026	B	C	C	C	C	P	C	NR	P	C	C
LDAR [83]	2026	B	C	C	C	C	C	C	NR	C	C	C
LiteRAG [84]	2026	T	C	C	C	P	C	P	P	P	C	C
LogicRAG [85]	2026	R	C	C	C	C	C	C	NR	C	P	C
LTRR [86]	2026	R	C	C	C	C	P	C	C	C	C	C
MA-RAG [87]	2026	S	C	C	C	C	C	C	C	C	C	C
MemGraphRAG [88]	2026	R	C	C	C	C	C	C	NR	C	P	C
METEORA [89]	2026	B	C	C	C	C	P	C	NR	C	C	C
MixRAGRec [90]	2026	R	C	C	C	C	C	C	NR	C	C	C
Mixture-of-RAG [91]	2026	B	C	C	C	C	C	C	NR	C	C	C
MoRE-RAG [92]	2026	R	C	C	C	C	C	C	C	P	C	C
NeocorRAG [93]	2026	B	C	C	C	C	C	C	P	C	C	C
ORDER [94]	2026	R	C	C	C	C	C	C	C	C	C	C
PROGRAM [95]	2026	R	C	C	C	P	C	C	NR	P	P	C
PruneRAG [96]	2026	S	C	C	C	C	C	C	P	C	C	C
QAFD-RAG [97]	2026	B	C	C	C	P	C	C	P	C	C	C
QR3AG [98]	2026	T	C	C	C	C	C	C	NR	P	P	C
QuDAR [99]	2026	R	C	C	C	C	P	C	P	C	C	C
Query-Aware Context Selection [100]	2026	B	C	C	C	C	P	C	C	P	C	C
R3-RAG (multimodal) [101]	2026	R	C	C	C	C	C	C	C	C	C	C
R3AG [102]	2026	R	C	C	C	C	C	C	NR	P	C	C
RAG-on-a-Diet [103]	2026	B	C	C	C	C	C	C	C	C	C	C
RAM [104]	2026	B	C	C	C	P	C	P	NR	P	P	C
ReaLM-Retrieve [105]	2026	T	C	C	C	C	C	C	P	C	C	C
Reflective RAG [106]	2026	S	C	C	C	P	P	C	NR	P	P	C
ReflectiveRAG [107]	2026	S	C	C	C	P	P	C	NR	P	P	C
RegionRAG [108]	2026	B	C	C	C	C	C	C	NR	C	C	C
REPAIR [109]	2026	R	C	C	C	C	C	C	P	C	C	C
Retriever Portfolios [110]	2026	R	C	C	C	P	C	C	P	P	P	C
RF-Mem [111]	2026	R	C	C	C	C	C	C	C	C	C	C
RouteRAG [112]	2026	R	C	C	C	C	C	C	NR	P	P	C
S-Path-RAG [113]	2026	R	C	C	C	C	C	C	C	P	C	C
S2G-RAG [114]	2026	S	C	C	C	C	C	C	P	C	C	C
SA-RAG [115]	2026	R	C	C	C	C	C	C	C	P	C	C
SARA [116]	2026	B	C	C	C	P	C	P	NR	P	P	C
SC-DRAG [117]	2026	T	C	C	C	C	C	C	NR	P	C	C
SC-RAG [118]	2026	B	C	P	C	C	C	C	NR	P	C	C
SegMem-RAG [119]	2026	R	C	C	C	C	P	C	NR	P	P	C
SmartChunk [120]	2026	B	C	C	C	C	C	C	NR	P	C	C
SPARC-RAG [121]	2026	B	C	C	C	C	C	C	C	C	C	C
SPARKLE [122]	2026	R	C	C	C	C	C	C	P	C	C	C
State-Aware RAG [123]	2026	B	C	C	C	C	C	C	NR	C	C	C
Structured Planning [124]	2026	R	C	C	C	C	P	C	NR	P	C	C
SUGAR-L [125]	2026	T	C	C	C	C	C	C	NR	P	C	C
TAdaRAG [126]	2026	R	C	P	C	C	P	C	NR	P	P	P
TGS-RAG [127]	2026	B	C	C	C	C	C	C	NR	C	NR	C
ThoraxAgent [128]	2026	B	C	C	C	C	C	C	NR	P	C	C
VimRAG [129]	2026	B	C	C	C	C	C	C	NR	P	C	C
X-Router [130]	2026	R	C	C	C	C	C	C	C	C	C	C
Zero-RAG [131]	2026	T	C	C	C	C	C	C	NR	P	C	C

S3 Illustrative control mechanisms

Table S3 compares examples that expose different control decisions. The phase column describes the behavior discussed here and need not list every phase in which a system operates. I denotes pre-retrieval; II, retrieval-time; III, post-retrieval evidence control; IV, reasoning, verification, and termination. The last column retains controlled quantities, measured costs, or implementation limits as stated in the comparisons. A dagger marks contextual examples outside Table S2. This table does not provide corpus-wide frequencies.

Table S3: Illustrative methods and their control decisions.

Year	Method	Phase(s)	Signal and action	Control target or resource note
2023	FLARE [1]	I	Future-token confidence. Trigger retrieval during generation and reformulate context	Retrieval frequency
2023	IRCoT [2]	II	Intermediate reasoning state. Interleave reasoning and search	Retrieval rounds
2023	Rewrite-Retrieve-Read [4]	I	Input query / rewriter state. Rewrite search query; reader feedback is used during training	Query quality
2023	SKR [5]	I	Estimated self-knowledge. Retrieve mainly for unknown questions	Unnecessary retrieval
2024	Adaptive-RAG [6]	I	Question complexity. Choose no-, one-, or multi-step retrieval	Strategy cost
2024	DRAGIN [8]	I, II	Real-time information need. Decide when and what to retrieve during generation	Retrieval timing
2024	EfficientRAG [9]	II	Iterative evidence state. Generate follow-up queries and filter with a smaller retriever	LLM calls
2024	RE-RAG [11]	III	Learned relevance confidence. Estimate usefulness and filter context	Context noise
2024	REAR [12]	III	Document relevance. Adapt external-knowledge utilization	Misleading evidence
2024	Self-RAG [14]	I, IV	Reflection tokens. Retrieve on demand and critique evidence/generation	Unnecessary retrieval
2025	Adaptive-k [18]	III	Similarity-score threshold. Choose a query-specific number of passages	Context tokens
2025	DioR [22]	I, III	Cognitive uncertainty + context scrutiny. Trigger retrieval and optimize retrieved context	Retrieval harm
2025	SeaKR [31]	I, II	Hidden-state self-aware uncertainty. Trigger, rerank, and choose reasoning strategy	Retrieval frequency
2025	SetR [132]†	III	Derived information requirements. Select a collectively sufficient passage set	Redundancy / coverage
2025	InfoGain-RAG [133]†	III	Generation-confidence change. Rerank/filter by document utility	Retrieval harm
2025	DynamicRAG [23]	III	Query + candidate documents. Adapt document order and query-specific k; generator-output quality is an RL training reward	Context size / noise
2025	GraphRAG-FI [134]†	III	Filter + logits-based integration. Filter graph evidence and balance with parametric reasoning	Graph noise / over-reliance
2025	MMAPG [135]†	II, IV	Planning-graph state. Choose next multimodal action and graph expansion	Search path errors
2025	FiSKE [25]	II, IV	Stateful clue–KG mappings. Adapt mapping, continue graph search, or fall back to CoT	LLM calls / redundant exploration
2025	EI-ARAG [24]	I	Contextualized token-embedding signal. Predict whether external retrieval is required without an extra LLM call	Controller overhead / retrieval need
2025	AMRAF [20]	I	Mentor-model generation assessment. Trigger or skip multi-level retrieval and regenerate with mentor feedback	Retrieval/regeneration cost
2026	GRIP [74]	I, II, IV	Generated control tokens + reasoning state. Retrieve, reformulate the query, integrate evidence, or terminate within decoding	Retrieval calls / loop length

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Table S3 continued

Year	Method	Phase(s)	Signal and action	Control target or resource note
2026	REPAIR [109]	II	Reasoning-plan step support. Selectively expand retrieval around pivotal under-supported reasoning steps	Candidate recall / expansion cost
2026	RAG-on-a-Diet [103]	IV	Entity + confidence state. Route each hop to the smallest sufficient model and stop by confidence/cap; cost-aware reward is used in training	Model cost / hop count
2026	ReaLM-Retrieve [105]	I, II	Step-level reasoning uncertainty. Intervene with retrieval only at reasoning steps predicted to benefit	Retrieval frequency / integration cost
2026	Cost-aware depth [55]	II, III	Query + retrieval state. Dynamically vary retrieved-list depth; cost-aware advantage is used in RL training	List length / latency / memory
2026	PruneRAG [96]	II, IV	Node confidence + decomposition-tree state. Expand, accept, or prune decomposition branches adaptively	Tree width/depth / redundant search
2026	LogicRAG [85]	I, II	Query-specific dependency DAG. Build/prune a reasoning DAG and retrieve by logical dependency	Retrieval/token redundancy
2026	DA-RAG [59]	II, III	Query attributes + graph topology. Dynamically extract attributed communities at multiple granularities	Subgraph/token/index cost
2026	TAdaRAG [126]	I, III	Task intent / domain template. Route extraction and construct a task-adaptive KG on the fly	Context noise / structure loss
2026	CompSelect [52]	III	Clue utility + minimum-sufficiency target. Extract, rerank, and adaptively truncate to a compact clue set	Reader tokens / latency
2026	SmartChunk [120]	II, III	Query-aware abstraction planner. Select retrieval granularity from document to sentence	Chunk granularity / cost
2026	GRAD [70]	III (decoding)	Task-dependent guidance utility. Compose and gate objective-specific decoding guidance at inference	Generation latency / evidence-use risk
2026	QR^{3}AG [98]	I	Query–response relevance threshold. Retrieve only when initial-response support is insufficient	Retrieval rate / redundancy
2026	INKER [77]	I, II	Query complexity + internal representation. Trigger retrieval and formulate a dual-signal query	Retrieval frequency / query quality
2026	ETC [64]	II	Entropy trend. Trigger retrieval on emerging uncertainty trends	Retrieval timing / frequency
2026	Flow-RAG [66]	II, III	Flow-score distribution + query structure. Set a query-specific subgraph pruning boundary	Graph context / latency
2026	SC-RAG [118]	III, IV	Fine-grained aspect evidence. Evidence-guided self-correction of internal–external conflict	Conflict / correction risk
2026	CRGS-RAG [136]†	III	Evidence support + conflict condition. Causal filtering and scenario-dependent knowledge fusion	Noise / knowledge conflict
2026	AIR-RAG [44]	II, III	LLM/retriever preference feedback. Select sentences, augment queries, and refine retrieval iteratively	Iterations / document refinement
2026	SARA [116]	III	Relevance + token budget. Retain selected text and compress remaining evidence	Input tokens
2026	QuDAR [99]	I	Score margins + relevance. Adapt sparse/dense and original/expanded query weights	Retriever mismatch / fixed hybrid weights
2026	SPARKLE [122]	II	Query + structured reasoning state. RL-trained proxy policy selects structured retrieval actions	Retrieval efficiency

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Table S3 continued

Year	Method	Phase(s)	Signal and action	Control target or resource note
2026	RouteRAG [112]	II, IV	Evolving reasoning + retrieved-evidence state. RL policy reasons, retrieves text/graph, or answers; quality/cost are training rewards	Text/graph retrieval cost
2026	AutoSearch [46]	IV	Intermediate-answer quality. Learn minimal sufficient search depth	Search steps
2026	FrugalRAG [67]	II, IV	Evolving search state. Reduce search depth according to question difficulty; frugality reward is used in training	Retrieval depth
2026	S2G-RAG [114]	IV	Sufficiency + structured gaps. Stop or retrieve missing evidence	Redundant search
2026	CIRAG [50]	II, III	Branch state + context need. Preserve evidence chains and expand granularity adaptively	Granularity / path errors
2026	UniversalRAG [137]†	I, II	Modality + query scope. Route to modality-specific corpus and granularity	Cross-modal mismatch
2026	DORA [63]	II	Question/image + reasoning and search history. Decide when to search during multimodal reasoning; quality/efficiency rewards are used in training	Search calls
2026	D^2-RAG [58]	I, III	Multi-dimensional uncertainty + context utility. Decide whether to retrieve, then adapt decoding	Retrieval harm
2026	SUGAR-L [125]	I, II, III	Semantic entropy. Choose no/single/multi-step retrieval; compress only when accumulated context overflows	Retrieval rounds / context length
2026	SC-DRAG [117]	I, II	Sentence-level verbal confidence. Progress, reflect/refine, or retrieve at sentence checkpoints	Retrieval timing / reflection cost
2026	SA-RAG [115]	II, IV	Evolving reasoning/graph state. Select Retrieve, Search, Complete, Correct, or Answer	Search actions / termination
2026	ACRA [38]	II, IV	Path score + depth-dependent threshold. Continue or terminate chain expansion using confidence-aware filtering	Chain depth / noisy branches
2026	NeocorRAG [93]	III	Evidence-chain support. Filter retrieved documents using generated evidence chains	Irrelevant documents / token use
2026	G^2-Reader [68]	I, II, IV	Evidence gaps revise the planning DAG; sufficiency or a round cap ends refinement	Readout budget; construction/inference tokens and time
2026	Failure is Feedback [65]	I, II, IV	Failure history selects re-anchoring, traversal scope, granularity, and LLM escalation	Traversal time; LLM calls and input/output tokens
2026	GAM-RAG [69]	II, III, IV	Sufficiency gates further propagation; uncertainty-weighted sentence memory updates after episodes	Indexing tokens; online latency and memory turns
2026	VimRAG [129]	II, III, IV	Graph importance allocates visual resolution and prunes redundant memory/actions	Visual and total tokens; action counts
2026	LAD-RAG [81]	I, II, IV	Agent selects semantic/symbolic tools, expands context, and checks whether to continue	Page count; retrieval calls, generated tokens and latency
2026	TGS-RAG [127]	II, III	Visited nodes rerank text; textual cues restore previously pruned graph connections	Retrieval/generation tokens; reused traversal history
2026	QAFD-RAG [97]	II	Query-aware flow gates expand an active subgraph until capacity/flow conditions are met	Explored subgraph size; indexing/query wall-clock time
2026	MemGraphRAG [88]	I, III	Empty retained schema/fact candidates trigger a passage-only retrieval fallback	Retrieval seconds; construction overhead
2026	Doc-REFRAG [62]	III	RL selector chooses which visual chunks to restore under a fixed expansion fraction	Restored visual tokens; time to first token and latency

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Table S3 continued

Year	Method	Phase(s)	Signal and action	Control target or resource note
2026	R^3AG [102]	I	Query-conditioned quality/utility representations select a retriever or no retrieval	Routing and retriever latency; fixed top-5 when retrieving
2026	AdaCache [39]	III	Confidence gates incremental use of cached, already-retrieved chunks	Context processed and time to first token; one retrieval call
2026	RF-Mem [111]	I, II	Probe similarity/entropy choose one-shot lookup or bounded iterative recollection	Retrieval latency and context tokens; fixed search caps
2026	Deliberative Searcher [61]	I, II, IV	Reasoning gaps trigger search/read actions; confidence is updated before answering	Training targets reliability; test-time rollout counts are measured
2026	GraphRAG-Router [73]	I	Query-conditioned hierarchy selects a GraphRAG pipeline and generator	Routing overhead and query cost; SFT/RL are offline
2026	SPARC-RAG [121]	II, III, IV	Shared memory supports iterative search and a learned continue/stop decision	Tokens/depth; fixed width of two
2026	AgenticTableRAG [43]	I, II, IV	Subqueries select table-QA tools; verification triggers retries	Tool calls and cost; heterogeneous table reasoning
2026	HALT [75]	IV	Per-hop verification stops search when all required claims match	Search and verification calls; gold claims are diagnostic only
2026	Cost-Aware Evidence Selection [56]	I, II, IV	Remaining budget informs source-tier selection and answer/continue actions	Simulated evidence-access costs, not measured market prices
2026	R^3-RAG [101]	I, III, IV	Modality dispersion gates reranking; diagnosed failures revise source queries	Fixed top-k, three-round cap; latency overhead reported
2026	CRAG [57]	II, III, IV	Agent selects retrieval, evaluation, and web-search tools before generation	Tool calls; no-tool decision or five-turn cap ends correction

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